GuitarTap Manual Test Checklist

Tester: _____ Date: _____ Build: _____

Platform(s) tested: [ ] iPhone [ ] iPad [ ] macOS

F — FFT Pipeline (Manual)

F21 — Live FFT renders in real time

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: App running with microphone permission granted.

- ☐ Launch the app (recording begins automatically)
- ☐ Verify the spectrum chart updates periodically (expected ~0.7 Hz — roughly once per 1.5 s)
- ☐ Confirm the display does not freeze permanently or show obviously stale data
- ☐ Note: the low update rate is by design — the large FFT window needed for high frequency resolution requires ~1.5 s of audio per frame
- ☐ **Pass** - [ ] **Fail** Notes: _____

F22 — Silence floor is plausible

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: App recording in a quiet environment.

- ☐ Launch the app in a quiet environment (recording begins automatically)
- ☐ Observe the noise floor level displayed in the spectrum
- ☐ Confirm the noise floor is visible and stable near −70 to −90 dB
- ☐ Confirm the floor is not jumping or showing false peaks
- ☐ **Pass** - [ ] **Fail** Notes: _____

F23 — Pure tone visible at correct frequency

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: A 220 Hz reference tone source is available (e.g. tuning app, online tone generator). In Settings, confirm the mode is set to a Guitar tap mode (not a Material tap mode), so that 220 Hz falls within the displayed frequency range.

- ☐ Launch the app (recording begins automatically)
- ☐ Play a 220 Hz tone near the device microphone
- ☐ Verify a peak appears in the spectrum chart near 220 Hz (within ±5 Hz)
- ☐ **Pass** - [ ] **Fail** Notes: _____

T — Tap Detection (Manual)

T9 — Tap detected on physical knock

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: App running and recording. Tap count set to 1. Acoustic measurement mode. Microphone (built-in or external) positioned close to the guitar body.

- ☐ Confirm spectrum is animating (recording starts automatically on launch)
- ☐ Knock the guitar top firmly once
- ☐ Verify the tap indicator flashes green
- ☐ Verify `Tap count: 1/1` appears in the status bar
- ☐ Verify the spectrum freezes and displays the captured response
- ☐ **Pass** - [ ] **Fail** Notes: _____

T10 — Minimum inter-tap interval enforced

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: App running and recording. Tap count set to 3.

- ☐ Knock the guitar twice in very rapid succession (<0.3 s apart)
- ☐ Verify only one tap is counted

- ☐ Confirm the second knock within the cooldown window is silently rejected
- ☐ **Pass** - [ ] **Fail** Notes: _____

T11 — Background noise does not false-trigger

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: App running and recording. Tap count set to 1. Device in a quiet room with no intentional taps.

- ☐ Leave recording with no intentional taps for 30 s
- ☐ Confirm the status message remains “Tap the guitar…” throughout
- ☐ Confirm the indicator does not flash green
- ☐ **Pass** - [ ] **Fail** Notes: _____

T12 — Threshold setting affects sensitivity

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: App running. Default threshold value set in Settings.

- ☐ Perform a light tap — confirm it is **not** detected at the default threshold
- ☐ Reduce the threshold using the Threshold slider until the threshold line on the chart is below the peak of the previous tap
- ☐ Perform the same light tap — confirm it **is** now detected
- ☐ Raise the threshold back above the tap peak — confirm light taps are missed again
- ☐ **Pass** - [ ] **Fail** Notes: _____

A — Averaging (Manual)

A3 — Multi-tap accumulation and progress bar

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: App running. Tap count set to 3.

- ☐ Knock the guitar once; verify progress bar advances to 1/3

- ☐ Knock again; verify progress bar advances to 2/3
- ☐ Knock a third time; verify progress bar reaches 3/3
- ☐ Confirm the averaged spectrum is displayed after the third tap
- ☐ **Pass** - [ ] **Fail** Notes: _____

A4 — Averaged spectrum is more consistent than a single tap

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: App running. A real guitar is available.

- ☐ Set tap count to 1; capture one tap; note the positions of the prominent peaks
- ☐ Tap **Reset**
- ☐ Set tap count to 5; capture five taps, deliberately varying tap position and force slightly between each
- ☐ Compare the two spectra — confirm the prominent peaks appear at the same frequencies (averaging reduces variation due to tap position and force, not spectral smoothness)
- ☐ **Pass** - [ ] **Fail** Notes: _____

A5 — New Tap clears previous capture

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: App running and recording. One or more taps have been captured (spectrum is frozen).

- ☐ Tap the **New Tap** button
- ☐ Verify the frozen spectrum clears
- ☐ Verify the status message returns to “Tap the guitar…”
- ☐ If a multi-tap sequence was in progress, confirm the progress bar also disappears
- ☐ **Pass** - [ ] **Fail** Notes: _____

S — Audio Stack (Manual)

S1 — First launch: microphone permission prompt

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: Permission must be in the "not determined" state. Toggling the microphone switch off in System Settings is **not** sufficient — that sets the status to "denied" and does not re-trigger the system prompt. To reset to "not determined": - **iOS/iPadOS:** Delete and reinstall the app, or use **System Settings → General → Transfer or Reset iPhone → Reset → Reset Location & Privacy**. - **macOS:** Run `tccutil reset Microphone` in Terminal (resets all apps), then relaunch GuitarTap.

- ☐ Launch GuitarTap
- ☐ Verify the OS permission dialog "GuitarTap would like to access the microphone" appears
- ☐ Confirm the app does not crash
- ☐ Confirm the app does not start recording before the user responds
- ☐ **Pass** - [ ] **Fail** Notes: _____

S2 — Permission denied: graceful degradation

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: Microphone permission previously denied (by tapping "Don't Allow" in S1, or by toggling off in System Settings). App running.

Note (macOS): Toggling the slider off in System Settings causes macOS to force-restart GuitarTap. The app checks permission on launch, so the alert appears automatically on restart — no need to press Start first.

- ☐ Launch GuitarTap (the app attempts to start recording automatically; the alert will appear at launch)
- ☐ Verify the app does not crash
- ☐ Verify recording does not start
- ☐ Confirm an alert titled "Microphone Access Required" appears
- ☐ Tap/click **Open Settings** and verify it navigates to System Settings → Privacy & Security → Microphone
- ☐ **Pass** - [ ] **Fail** Notes: _____

S3 — Permission granted after denial

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: Microphone permission was previously denied.

- ☐ Open **System Settings → Privacy & Security → Microphone → GuitarTap** and enable it
- ☐ **iOS/iPadOS:** Return to GuitarTap (no relaunch needed on iOS 14+) — recording restarts automatically
- ☐ **macOS:** GuitarTap is force-restarted automatically; recording begins on relaunch
- ☐ Verify recording begins and the spectrum starts animating
- ☐ Verify the green running indicator appears
- ☐ **Pass** - [ ] **Fail** Notes: _____

S4 — Microphone-in-use indicator active

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: App running on a real device.

- ☐ Launch the app (recording begins automatically)
- ☐ **iOS/iPadOS:** Confirm the orange microphone indicator appears in the status bar
- ☐ **macOS:** Confirm the orange microphone indicator appears in the menu bar (top-right area); note that it persists until the app is quit — this is normal macOS behavior
- ☐ **Pass** - [ ] **Fail** Notes: _____

S5 — Session interrupted by incoming call

Platforms: iOS **Priority:** High

Preconditions: App recording on a real iPhone.

- ☐ Confirm spectrum is animating (recording starts automatically on launch)
- ☐ Have another phone call the device

- ☐ Let the call ring for 5 s then decline (do not answer)
- ☐ While the call rings, verify recording stops cleanly (no crash, no freeze)
- ☐ After declining, verify the app is in a consistent state
- ☐ **Pass** - [ ] **Fail** Notes: _____

S6 — Session interrupted by another audio app

Platforms: iOS · iPadOS **Priority:** Medium

Preconditions: App recording on iPhone or iPad.

Note: Does not apply to macOS — CoreAudio allows simultaneous audio access by multiple apps, so a playback app does not interrupt GuitarTap's microphone input.

- ☐ Confirm spectrum is animating (recording starts automatically on launch)
- ☐ Swipe to the Home screen
- ☐ Open Music or Spotify and start playback
- ☐ Return to GuitarTap
- ☐ Verify GuitarTap is in a stopped state
- ☐ Verify the app recovers when focus returns
- ☐ **Pass** - [ ] **Fail** Notes: _____

S7 — Input device selection (macOS)

Platforms: macOS **Priority:** High

Preconditions: Two microphone sources available on the Mac (e.g. built-in + USB headset).

- ☐ Launch GuitarTap (recording begins automatically)
- ☐ Open GuitarTap **Settings → Audio Input & Calibration → Audio Input Device** and select the alternative device
- ☐ Within 2 s, verify the spectrum updates to reflect audio from the new device
- ☐ Confirm no crash occurs
- ☐ **Pass** - [ ] **Fail** Notes: _____

S8 — External device plug/unplug (macOS)

Platforms: macOS **Priority:** Medium

Preconditions: App recording on macOS.

- ☐ Plug in a USB microphone
- ☐ Confirm it appears in the device list in Settings
- ☐ Select it; confirm recording continues
- ☐ Unplug the USB microphone
- ☐ Verify the app does not crash
- ☐ Verify the device list updates
- ☐ Verify recording either falls back to the built-in mic or stops cleanly
- ☐ **Pass** - [ ] **Fail** Notes: _____

S9 — 48 kHz sample rate accepted

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: App recording on any platform.

- ☐ Launch the app (recording begins automatically)
- ☐ Tap **Metrics** to open the metrics overlay
- ☐ Verify **Sample Rate** shows “48000 Hz” (or the device’s actual rate)
- ☐ Confirm the value is not “0 Hz”, “—”, or empty
- ☐ **Pass** - [ ] **Fail** Notes: _____

S10 — Non-48 kHz device accepted

Platforms: macOS **Priority:** Low

Preconditions: A microphone whose sample rate can be changed in **Audio MIDI Setup** (Applications → Utilities) — the built-in microphone works.

- ☐ Open Audio MIDI Setup and set the device to 44100 Hz

- ☐ Launch GuitarTap and select that device in **GuitarTap Settings → Audio Input & Calibration → Audio Input Device**
- ☐ Confirm recording starts automatically
- ☐ Verify the spectrum chart appears and animates normally
- ☐ Confirm **Sample Rate** in Metrics reads “44100 Hz”
- ☐ While still recording, switch the device back to 48000 Hz in Audio MIDI Setup
- ☐ Verify recording stops briefly then restarts automatically
- ☐ Confirm **Sample Rate** in Metrics now reads “48000 Hz”
- ☐ Confirm the app does not crash or show an error dialog
- ☐ **Pass** - [ ] **Fail** Notes: _____

C — Chart Interaction (Manual)

C9 — Pinch-to-zoom narrows frequency range

Platforms: iOS · iPadOS **Priority:** High

Preconditions: App running with a frozen or live spectrum.

- ☐ Place two fingers near the centre of the spectrum chart
- ☐ Pinch inward (bring fingers together)
- ☐ Verify the frequency axis zooms in and the visible range narrows
- ☐ Confirm axis tick labels update accordingly
- ☐ **Pass** - [ ] **Fail** Notes: _____

C10 — Spread-to-zoom widens frequency range

Platforms: iOS · iPadOS **Priority:** High

Preconditions: Frequency axis is already zoomed in (range < 1000 Hz).

- ☐ Place two fingers near the centre of the chart
- ☐ Spread fingers apart
- ☐ Verify the frequency axis zooms out and the range widens

- ☐ Confirm the range cannot exceed 0–5000 Hz (linear scale) or 20–5000 Hz (logarithmic scale)
- ☐ **Pass** - [ ] **Fail** Notes: _____

C11 — Drag pans frequency axis

Platforms: iOS · iPadOS **Priority:** High

Preconditions: Frequency axis is zoomed in to a sub-range.

- ☐ Press and drag the chart to the left; verify frequency axis shifts left
- ☐ Release; confirm position is maintained and zoom span has not changed
- ☐ Drag back to the right; verify frequency axis shifts right
- ☐ **Pass** - [ ] **Fail** Notes: _____

C12 — Scroll wheel pans magnitude axis (macOS)

Platforms: macOS **Priority:** High

Preconditions: App running on macOS with a trackpad or scroll wheel.

- ☐ Position cursor over the chart
- ☐ Scroll up (no modifier keys)
- ☐ Verify dB axis shifts upward (higher dB values come into view)
- ☐ Confirm frequency axis does not change
- ☐ **Pass** - [ ] **Fail** Notes: _____

C13 — Shift+scroll pans frequency axis (macOS)

Platforms: macOS **Priority:** High

Preconditions: App running on macOS.

- ☐ Hold **Shift** and scroll left/right
- ☐ Verify frequency axis pans left or right
- ☐ Confirm dB axis does not change

☐ **Pass** - [ ] **Fail** Notes: _____

C14 — Cmd+scroll zooms both axes (macOS)

Platforms: macOS **Priority:** High

Preconditions: App running on macOS. Cursor positioned over a peak.

- ☐ Hold **Cmd** and scroll up (zoom in)
- ☐ Verify both frequency and dB axes zoom in simultaneously
- ☐ Confirm the point under the cursor remains approximately fixed
- ☐ **Pass** - [ ] **Fail** Notes: _____

C15 — Option+scroll zooms frequency only (macOS)

Platforms: macOS **Priority:** Medium

Preconditions: App running on macOS.

- ☐ Hold **Option** and scroll up
- ☐ Verify frequency axis zooms in
- ☐ Confirm dB axis does not change
- ☐ **Pass** - [ ] **Fail** Notes: _____

C16 — Ctrl+scroll zooms magnitude only (macOS)

Platforms: macOS **Priority:** Medium

Preconditions: App running on macOS.

- ☐ Hold **Ctrl** and scroll up
- ☐ Verify dB axis zooms in
- ☐ Confirm frequency axis does not change
- ☐ **Pass** - [ ] **Fail** Notes: _____

C17 — Zoom anchors to cursor position (macOS)

Platforms: macOS **Priority:** Medium

Preconditions: App with a visible peak at a known frequency (e.g. 200 Hz).

- ☐ Position the cursor exactly over the 200 Hz peak
- ☐ Hold **Cmd** and scroll in to zoom
- ☐ Verify the 200 Hz peak remains under the cursor as the axes zoom in
- ☐ **Pass** - [ ] **Fail** Notes: _____

C18 — Reset axes to defaults (macOS/iOS/iPadOS)

Platforms: macOS · iOS · iPadOS **Priority:** Medium

Preconditions: Frequency and/or dB axes have been panned or zoomed away from defaults.

macOS: - [ ] Right-click the chart area to open the context menu - [ ] Select **Reset Axes to Defaults** - [ ] Verify both axes return to the saved default values (as stored in Settings)

iOS/iPadOS: - [ ] Tap the **info icon** on the chart - [ ] Tap **Reset Axes to Defaults** - [ ] Verify both axes return to the saved default values (as stored in Settings)

- ☐ **Pass** - [ ] **Fail** Notes: _____

C19 — Settings axis ranges preserved on restart

Platforms: iOS · iPadOS · macOS **Priority:** Low

Preconditions: App running on any platform.

- ☐ Open Settings and set non-default values for Min Frequency, Max Frequency, Min dB, and Max dB (e.g. 100–500 Hz, −100 to −20 dB)
- ☐ Close Settings and confirm the chart reflects the new ranges
- ☐ Quit the app fully (swipe up on iOS/iPadOS; Cmd+Q on macOS)
- ☐ Relaunch GuitarTap
- ☐ Verify the chart opens with the frequency and dB ranges set in Settings
- ☐ Note: gesture-based pan/zoom is intentionally not persisted and will revert to Settings values on restart

☐ **Pass** - [ ] **Fail** Notes: _____

D — Draggable Peak Annotations (Manual)

D11 — Annotation draggable on screen

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: A measurement has been captured and peak annotations are visible.

- ☐ Press and drag (iOS/iPadOS) or click-and-drag (macOS) a peak label
- ☐ Drag it to a new position
- ☐ Release
- ☐ Verify the label is at the new position and stays there
- ☐ **Pass** - [ ] **Fail** Notes: _____

D12 — Annotation position persists after zoom/pan

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: D11 completed; a label has been dragged to a non-default position.

- ☐ Pinch to zoom (iOS/iPadOS) or scroll with Command/Option/Control modifier (macOS) to zoom the chart
- ☐ Pan the chart
- ☐ Verify the annotation label stays attached to its peak at the correct chart position
- ☐ Confirm the label does not jump back to the default position
- ☐ **Pass** - [ ] **Fail** Notes: _____

D13 — Annotation reset to default position

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: A label has been dragged to a non-default position (D11).

- ☐ iOS/iPadOS: Double-tap the label; macOS: right-click and choose “Reset Position”

- ☐ Verify the label snaps back to its default (zero-offset) position relative to the peak marker
- ☐ **Pass** - [ ] **Fail** Notes: _____

D14 — Annotations hidden in None visibility mode

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: A measurement is captured and annotations are visible.

- ☐ Tap the **Annotations** button until the icon/tooltip indicates “None”
- ☐ Verify all peak annotation labels disappear from the chart
- ☐ **Pass** - [ ] **Fail** Notes: _____

D15 — Selected mode shows only Results-panel-selected labels

Platforms: iOS · iPadOS · macOS **Priority:** Low

Preconditions: Multiple peaks are visible and at least one peak is not selected in the Results panel.

- ☐ In the Results panel, confirm some peaks are selected and some are not
- ☐ Cycle the Annotations button to the **Selected** mode
- ☐ Verify only the labels for Results-panel-selected peaks are visible
- ☐ Verify labels for unselected peaks are hidden
- ☐ **Pass** - [ ] **Fail** Notes: _____

D16 — Guitar mode auto-selects one peak per mode on freeze

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: App in guitar measurement mode. A real guitar (or tone source covering multiple mode frequencies) available.

- ☐ Capture a tap so the spectrum freezes
- ☐ Open the Results panel

- ☐ Verify that at most one peak per mode (Air, Top, Back, Dipole, Ring Mode, Upper Modes) is selected (checkmark shown)
- ☐ Verify that if two mode ranges overlap and a peak falls in both, the peak is assigned to the lower-frequency mode
- ☐ Verify that peaks with mode "Unknown" are **not** auto-selected (no checkmark)
- ☐ Confirm the **Reset to Auto** wand button (⌛) in the Results panel header is disabled (already in auto state)
- ☐ **Pass** - [ ] **Fail** Notes: _____

D17 — Peak Min recalculates frozen guitar spectrum; selections update correctly

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: A guitar tap has been captured and the spectrum is frozen with at least two annotated peaks.

Part A — Auto state (no manual changes): - [ ] Without touching any selection checkboxes, move the Peak Min slider to the right (raise threshold) - [ ] Verify peaks that fall below the new threshold disappear from the chart and Results panel - [ ] Verify the remaining auto-selected peaks update (wand button stays disabled — still in auto state) - [ ] Move the slider back left; verify the previously hidden peaks reappear and auto-selection re-runs

Part B — Manual state: - [ ] Toggle any peak's checkmark in the Results panel (select or deselect it) - [ ] Confirm the **Reset to Auto** wand button is now enabled - [ ] Move the Peak Min slider - [ ] Verify peaks that survive the threshold carry their selection state forward (a peak you selected stays selected; one you deselected stays deselected) - [ ] Verify peaks removed by the threshold are simply gone (no crash, no stale IDs)

- ☐ **Pass** - [ ] **Fail** Notes: _____

D18 — Reset to Auto restores mode-priority selection

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: A guitar tap is frozen and the user has made at least one manual selection change (D17 Part B).

- ☐ Confirm the **Reset to Auto** wand button (⌛) in the Results panel header is enabled
- ☐ Tap/click the wand button

- ☐ Verify the selection reverts to auto-selection (one best peak per mode, lowest-first)
- ☐ Verify the wand button becomes disabled again
- ☐ Verify moving the Peak Min slider now re-runs auto-selection (not carry-forward)
- ☐ **Pass** - [ ] **Fail** Notes: _____

R — Persistence & Export (Manual)

R7 — Saved measurement persists across relaunch

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: At least one tap has been captured.

- ☐ Tap **Save**; enter a name (e.g. “Test spruce”); confirm
- ☐ Force-quit the app (swipe up from app switcher or Cmd+Q on macOS)
- ☐ Relaunch GuitarTap
- ☐ Open **Measurements**
- ☐ Verify the saved measurement appears in the list with the correct name and date
- ☐ **Pass** - [ ] **Fail** Notes: _____

R8 — Export Measurement round-trip — export and re-import matches original

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: A measurement with peaks is saved.

- ☐ Open the Measurements list; right-click (macOS) or long-press (iOS/iPadOS) the row and choose **View Details**; note the name, date, location/notes, and all peaks in the Results panel
- ☐ Right-click (macOS) or long-press (iOS/iPadOS) the same row and choose **Save Measurement to Disk…** (macOS) or **Export Measurement** (iOS/iPadOS); save the `.guitartap` file
- ☐ Import the file using the **Import** toolbar button (macOS) or ellipsis → **Import Measurement** (iOS/iPadOS)

- ☐ Right-click (macOS) or long-press (iOS/iPadOS) the newly imported row and choose **View Details**
- ☐ Verify the name, date, and location/notes match the original
- ☐ Verify the Results panel shows the same peaks with the same frequencies, magnitudes, and mode assignments
- ☐ Verify the spectrum chart appearance matches the original
- ☐ **Pass** - [ ] **Fail** Notes: _____

R9 — Export PDF renders correctly

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: A measurement with at least one peak is saved.

- ☐ Open the measurement detail view
- ☐ Tap **Export → PDF**
- ☐ Open the PDF in Preview (macOS) or Files (iOS)
- ☐ Verify the PDF contains the spectrum chart image
- ☐ Verify the PDF contains a table of peaks (frequency, dB, Q, mode)
- ☐ Verify the PDF contains measurement metadata (name, date, guitar type)
- ☐ Confirm there are no blank pages or missing images
- ☐ **Pass** - [ ] **Fail** Notes: _____

R10 — Cross-device transfer via export/import

Platforms: iOS · iPadOS · macOS **Priority:** Low

> **Note:** No automatic iCloud sync — cross-device transfer requires manual export/import.

Preconditions: Two devices with GuitarTap installed. A measurement is saved on device A.

- ☐ On device A: choose **Export Measurement** (iOS/iPadOS, or macOS for share sheet) or **Save Measurement to Disk…** (macOS for save to disk); send the `.guitartap` file to device B via AirDrop, Files/iCloud Drive, or email; AirDrop will route the file directly to GuitarTap on the receiving device

- ☐ On device B: import the file using **Import** toolbar button (macOS) or ellipsis → **Import Measurement** (iOS/iPadOS); legacy `.json` files are also accepted
- ☐ Verify the measurement appears in the list on device B
- ☐ **Pass** - [ ] **Fail** Notes: _____

CP — Comparison Mode (Manual)

CP1 — Enter and exit compare mode

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: At least 2 saved guitar measurements exist in the Measurements list.

- ☐ Open the **Measurements** list
- ☐ Tap/click **Compare**
- ☐ Verify the list enters selection mode (checkboxes appear, Import and Delete All are disabled)
- ☐ Tap/click **Cancel** (or the Compare button again if applicable)
- ☐ Verify the list returns to normal mode
- ☐ **Pass** - [ ] **Fail** Notes: _____

CP2 — Plate/brace measurements not selectable in compare mode

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: The Measurements list contains at least one plate or brace measurement and at least one guitar measurement.

- ☐ Enter compare mode (tap **Compare**)
- ☐ Tap a plate or brace measurement row
- ☐ Verify it does not become selected (no checkmark appears)
- ☐ Verify the **Compare Selected** button does not activate for plate/brace-only selection
- ☐ **Pass** - [ ] **Fail** Notes: _____

CP3 — Compare Selected loads overlay on main chart

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: At least 2 saved guitar measurements with spectrum snapshots exist.

- ☐ Enter compare mode
- ☐ Select exactly 2 guitar measurements
- ☐ Verify **Compare Selected (2)** button is now enabled
- ☐ Tap/click **Compare Selected (2)**
- ☐ Verify the Measurements list closes and the main view reappears
- ☐ Verify the main chart shows 2 coloured curves (not the normal red curve)
- ☐ Verify a legend is visible (either on the chart or in the status bar area)
- ☐ Verify the status bar shows “Comparing 2 measurements” and “Press New Tap to exit comparison”
- ☐ **Pass** - [ ] **Fail** Notes: _____

CP4 — Compare up to 5 measurements

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: At least 5 saved guitar measurements with spectrum snapshots exist.

- ☐ Enter compare mode and select 5 guitar measurements
- ☐ Confirm **Compare Selected (5)** is enabled
- ☐ Tap/click **Compare Selected (5)**
- ☐ Verify 5 coloured curves appear on the chart, each in a distinct colour (blue, orange, green, purple, teal)
- ☐ **Pass** - [ ] **Fail** Notes: _____

CP5 — New Tap exits comparison mode

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: Comparison mode is active (CP3 completed).

- ☐ Confirm the chart shows multiple coloured comparison curves
- ☐ Press/click **New Tap**
- ☐ Verify the chart returns to the normal single red curve (or live spectrum)
- ☐ Verify the status bar returns to tap detection state
- ☐ Verify the Results panel no longer shows “Comparing N measurements”
- ☐ **Pass** - [ ] **Fail** Notes: _____

CP6 — Comparison mode disables Save and Annotations

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: Comparison mode is active.

- ☐ Verify the **Save** button is visually disabled (greyed out)
- ☐ Verify the **Annotations** button is visually disabled
- ☐ Attempt to click/tap Save — confirm nothing happens and no sheet appears
- ☐ Press **New Tap** to exit; verify both buttons re-enable
- ☐ **Pass** - [ ] **Fail** Notes: _____

CP7 — Results panel during comparison

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: Comparison mode is active.

- ☐ Open the **Results** panel
- ☐ Verify the header badge shows “Comparison” (purple) instead of the measurement type (e.g. “Classical”)
- ☐ Verify the peaks list shows “Comparing N measurements” instead of individual peak rows
- ☐ Verify Ring-Out Time and Tap Tone Ratio sections are **not** shown
- ☐ Verify **Export PDF Report** button is **not** shown
- ☐ Verify **Export Spectrum** button **is** shown

- ☐ **Pass** - [ ] **Fail** Notes: _____

CP8 — Export Spectrum in comparison mode

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: Comparison mode is active with 2+ measurements.

- ☐ Open the Results panel (or use the export action from the main view)
- ☐ Tap/click **Export Spectrum**
- ☐ Open the saved PNG image
- ☐ Verify the image shows all comparison curves (one per selected measurement)
- ☐ Verify the image shows a legend with coloured line swatches and measurement labels (not “Guitar Modes”)
- ☐ Verify no peak annotation boxes appear on the image
- ☐ Verify the header says “Comparing N measurements” (not “Detected Peaks: 0”)
- ☐ **Pass** - [ ] **Fail** Notes: _____

CP9 — Auto dB in comparison mode

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: Comparison mode is active. The comparison curves are partially off-screen.

- ☐ Tap/click the **Auto dB** button
- ☐ Verify the dB axis adjusts so all comparison curves are visible within the chart bounds
- ☐ Confirm no curve is clipped at the top or bottom of the chart
- ☐ **Pass** - [ ] **Fail** Notes: _____

CP10 — Cursor/crosshair works during comparison

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: Comparison mode is active with 2+ curves visible.

- ☐ Move the cursor over the chart (macOS) or enable and drag the crosshair (iOS/iPadOS)
- ☐ Verify a vertical cursor line appears that snaps to the nearest comparison curve
- ☐ Verify a frequency and magnitude readout is displayed
- ☐ Confirm no crash occurs
- ☐ **Pass** - [ ] **Fail** Notes: _____

U — Cross-Platform UI (Manual)

U1 — All primary controls present on iPhone

Platforms: iOS **Priority:** High

Preconditions: iPhone with compact horizontal size class (iPhone SE, iPhone 15, etc.).

- ☐ Launch GuitarTap on iPhone in portrait
- ☐ Verify nav bar left: Results
- ☐ Verify nav bar right: Metrics, Settings
- ☐ Verify header button bar: Auto dB, Crosshair, Annotations, Save, Measurements (icon-only bordered buttons)
- ☐ Rotate to landscape
- ☐ Verify nav bar left: Results, Auto dB, Crosshair
- ☐ Verify nav bar right: Annotations, Save, Measurements, Metrics, Settings
- ☐ Confirm no button is clipped, hidden, or missing in either orientation
- ☐ **Pass** - [ ] **Fail** Notes: _____

U2 — All primary controls present on iPad

Platforms: iPadOS **Priority:** High

Preconditions: iPad in landscape or portrait orientation.

- ☐ Launch GuitarTap on iPad

- ☐ Verify the two-row header layout shows full-label buttons
- ☐ Confirm both rows are fully visible without horizontal scrolling
- ☐ Confirm all buttons have text labels
- ☐ **Pass** - [ ] **Fail** Notes: _____

U3 — All primary controls present on macOS (wide window)

Platforms: macOS **Priority:** High

Preconditions: macOS, window at ≥900 px wide.

- ☐ Launch GuitarTap
- ☐ Verify the single-row button layout shows (right-justified): Auto dB, Annotations, Save, Measurements, Metrics, Settings
- ☐ Confirm all buttons have full text labels and are visible in one row
- ☐ **Pass** - [ ] **Fail** Notes: _____

U4 — macOS icon-only layout at narrow window

Platforms: macOS **Priority:** Medium

Preconditions: macOS.

- ☐ Drag the window to a narrow width (~500 px)
- ☐ Verify `ViewThatFits` switches to the icon-only two-row layout
- ☐ Confirm no button labels are truncated
- ☐ Confirm no button falls off-screen
- ☐ **Pass** - [ ] **Fail** Notes: _____

U5 — Status bar correct on iPhone

Platforms: iOS **Priority:** Medium

Preconditions: App running on iPhone.

- ☐ Launch the app (recording begins automatically)

- ☐ Verify the compact status bar (28 px tall) shows the tap indicator
- ☐ Verify it shows the dB reading
- ☐ Verify it shows the running dot indicator
- ☐ **Pass** - [ ] **Fail** Notes: _____

U6 — Status bar correct on iPad/macOS

Platforms: iPadOS · macOS **Priority:** Medium

Preconditions: App running on iPad or macOS.

- ☐ Launch the app (recording begins automatically); perform at least one tap
- ☐ Verify the full status bar shows the progress bar
- ☐ Verify it shows the “Tap Detected!” label
- ☐ Verify it shows the bin count
- ☐ Verify it shows the frozen indicator
- ☐ **Pass** - [ ] **Fail** Notes: _____

U7 — Settings sheet opens, changes persist

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: App running on any platform.

- ☐ Tap **Settings**
- ☐ Change the threshold value
- ☐ Dismiss the settings sheet
- ☐ Reopen **Settings**
- ☐ Verify the threshold value matches what was entered
- ☐ **Pass** - [ ] **Fail** Notes: _____

U8 — Results sheet opens and closes

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: At least one tap has been captured.

- ☐ Tap **Results**
- ☐ Verify the peaks table appears
- ☐ Dismiss the sheet
- ☐ Confirm the sheet closes cleanly
- ☐ **Pass** - [ ] **Fail** Notes: _____

U9 — Save sheet accepts name and saves

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: At least one tap has been captured.

- ☐ Tap **Save**
- ☐ Type a name for the measurement
- ☐ Confirm the save
- ☐ Open **Measurements**
- ☐ Verify the new measurement appears in the list
- ☐ **Pass** - [ ] **Fail** Notes: _____

U10 — Measurements list shows saved items

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: At least one measurement has been previously saved.

- ☐ Open **Measurements**
- ☐ Verify saved items are listed with name and date
- ☐ **Pass** - [ ] **Fail** Notes: _____

U11 — Dark mode: no invisible text

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: System dark mode enabled.

- ☐ Launch GuitarTap
- ☐ Review the main view — confirm all text is legible
- ☐ Review the status bar — confirm all text is legible
- ☐ Open and review the settings sheet — confirm all text is legible
- ☐ Open and review the measurement detail view — confirm all text is legible
- ☐ Confirm chart axis labels and peak annotations remain readable
- ☐ Confirm no white-on-white or black-on-black combinations
- ☐ **Pass** - [ ] **Fail** Notes: _____

U12 — Dynamic Type: large text accessible

Platforms: iOS · iPadOS **Priority:** Low

Preconditions: Accessibility text size set to the largest option via **Settings → Accessibility → Display & Text Size → Larger Text** (drag slider to maximum).

- ☐ Launch GuitarTap
- ☐ Inspect all buttons in the header and toolbar — confirm no label is clipped or truncated
- ☐ Open the **Settings** sheet and inspect all labels and controls
- ☐ Open the **Measurements** list and inspect row labels
- ☐ Verify all labels and controls remain usable
- ☐ Confirm the layout may reflow but must not hide or overlap any interactive element
- ☐ **Pass** - [ ] **Fail** Notes: _____

K — Regression Smoke Tests (Manual)

K1 — App launches without crash

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: Device restarted (cold launch).

- ☐ Tap the GuitarTap icon
- ☐ Verify the main view appears within 3 s
- ☐ Confirm no crash occurs
- ☐ Confirm no "GuitarTap quit unexpectedly" dialog appears
- ☐ **Pass** - [ ] **Fail** Notes: _____

K2 — Chart renders with a saved measurement

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: At least one measurement has been saved using the **Save** button during a previous tap tone session (see R8).

- ☐ Export the saved measurement to a `.guitartap` file using **Export Measurement** (iOS/iPadOS share sheet) or **Save Measurement to Disk…** (macOS)
- ☐ Re-import it using **Import Measurement** (iOS/iPadOS) or the **Import** toolbar button (macOS)
- ☐ Open the imported measurement from the Measurements list
- ☐ Verify the spectrum chart renders with peaks annotated
- ☐ Confirm no blank chart or decode error
- ☐ **Pass** - [ ] **Fail** Notes: _____

K3 — Recording starts automatically on launch

Platforms: iOS · iPadOS · macOS **Priority:** High

Preconditions: Microphone permission granted.

- ☐ Launch the app

- ☐ Verify the green running indicator appears within 3 s
- ☐ Confirm the app does not freeze or crash
- ☐ **Pass** - [ ] **Fail** Notes: _____

K4 — Memory stable during extended recording

Platforms: iOS · iPadOS · macOS **Priority:** Medium

Preconditions: Xcode Instruments available. App running on a physical device.

- ☐ Profile the app with the Allocations instrument in Xcode
- ☐ Launch the app (recording begins automatically)
- ☐ Record continuously for 2 minutes
- ☐ Verify memory usage grows during FFT initialisation and then plateaus
- ☐ Confirm there is no unbounded growth over the 2-minute period
- ☐ Confirm net growth is less than 10 MB from the plateau
- ☐ **Pass** - [ ] **Fail** Notes: _____

K5 — App returns from background correctly

Platforms: iOS · iPadOS **Priority:** High

Preconditions: App recording on iPhone or iPad.

- ☐ Confirm spectrum is animating (recording starts automatically on launch)
- ☐ Press the Home button (or swipe up to go home)
- ☐ Wait 10 s
- ☐ Return to GuitarTap by tapping its icon
- ☐ Verify the app is either still recording (indicator green) or in a clean stopped state
- ☐ Confirm the UI is not frozen
- ☐ Confirm no crash has occurred
- ☐ **Pass** - [ ] **Fail** Notes: _____

K6 — Rotation does not break layout

Platforms: iOS · iPadOS **Priority:** Medium

Preconditions: App running on iPhone or iPad.

- ☐ Hold the device in portrait; confirm the spectrum is visible and animating (recording starts automatically on launch)
- ☐ Rotate the device to landscape
- ☐ Verify the chart, header buttons, and status bar reflow without truncation or overlap
- ☐ Rotate back to portrait
- ☐ Verify the layout is still correct
- ☐ **Pass** - [ ] **Fail** Notes: _____

Summary

Area	Total Manual Tests	Pass	Fail	Skipped
F — FFT Pipeline	3			
T — Tap Detection	4			
A — Averaging	3			
S — Audio Stack	10			
C — Chart Interaction	11			
D — Draggable Annotations	8			
R — Persistence & Export	4			
CP — Comparison Mode	10			
U — Cross-Platform UI	12			
K — Regression Smoke	6			
Total	**71**			

Overall result: Pass / Fail / Partial **Sign-off:** _____